Experiment :1

Develop a LaTeX script to create a simple document that consists of 2 sections [Section1, Section2], and a paragraph with dummy text in each section. And also include header [title of document] and footer [institute name, page number] in the document.

```
\documentclass[12pt,a4paper]{article}
\usepackage[left=2cm,right=2cm,top=2cm,bottom=2cm]{geometry}
\usepackage{fancyhdr}
\begin{document}
% Set the page style to "fancy"...
\pagestyle{fancy}
\title{TJIT}

\fancyhf{} % clear existing header/footer entries
% We don't need to specify the O coordinate
\fancyhead{} % clear all header fields
\fancyhead[R]{TJIT}
\fancyfoot{}  % clear all footer fields
\fancyfoot[LO,CE]{TJIT,Bangalore}

\fancyfoot[R]{\thepage}
\maketitle
```

\section{About TJIT?}
T John Institute of Technology (TJIT) is a private Engineering college located in Bangalore,Karnataka,India.It was established in 2006 by Thomas P John and is part of T John Group of Institutions.The Institute offers undergraduate and post graduate programmes and is affiliated to Visvesvaraya Technological University(VTU) and is approved by All India Council for Technical Education (AICTE).

\section{History of TJIT}
T John Institute of Technology is a part of T John Group of Institutions.Since a humble beginning in 1993 the T John group has expanded to include new department and blocks and spread over an area of 20 acres.TJIT was set up in the year 2006.Starting with just two courses B.E Computer Science Engineering and B.E Electronics and Communication Engineering,today the college has expanded into a provider of Technical Education offering B.E degrees in six streams of Engineering as well as two post graduate courses.\\

The department of Information Science Engineering and the department of Telecommunication Engineering were established in the year 2007.In the same year the Institute also started the Business Administration department that offers the MBA programme under VTU.In 2009 ,the first batch of Mechanical Engineering students joined the Institute to pursue B.E in Mechanical Engineering under the newly established department of Mechanical Engineering.Continuing its growth in 2010,TJIT started MCA programme under VTU.

\section{Departments}

```
\begin{itemize}
\item Department of Computer Science Engineering
\item Department of CSE(IOT and Cybersecurity including Block chain Technology)
\item Department of Data Science Engineering
\item Department of Information Science Engineering
\item Department of Electronics and Communication Engineering
\item Department of Mechanical Engineering
\item Department of Civil Engineering
\item Department of Humanities and Sciences
```

\item Department of Business Administration
\item Department of Computer Applications
\end{itemize}
\section{Facilities and Extracurricular Activities}

\begin{itemize}
\item Cafeteria
\item Digital Library
\item Auditorium
\item Laboratory
\item Hostel
\item Sports
\item Seminar
\item Conference
\item Students Chapter

\end{itemize}

\end{document}

OUTPUT:

[Type here]

TJIT

February 26, 2025

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3 Departments

- Department of Computer Science Engineering
- Department of CSE(IOT and Cybersecurity including Block chain Technology)
- Department of Data Science Engineering
- Department of Information Science Engineering
- Department of Electronics and Communication Engineering
- Department of Mechanical Engineering
- Department of Civil Engineering

- Department of Humanities and Sciences
- Department of Business Administration
- Department of Computer Applications

4 Facilities and Extracurricular Activities

- Cafeteria
- Digital Library
- Auditorium
- Laboratory
- Hostel
- Sports
- Seminar
- Conference
- Students Chapter

Experiment 2:

Develop a LaTeX script to create a document that displays the sample Abstract/Summary

```latex
\documentclass[10pt,a4paper]{article}
\usepackage[utf8]{inputenc}
\usepackage{amsmath}
\usepackage{amsfonts}
\usepackage{amssymb}
\usepackage[left=3cm,right=3cm,top=2cm,bottom=2cm]{geometry}
%\usepackage{lipsum}

\begin{document}
\thispagestyle{plain}

\begin{center}
   \Large
   \textbf{A Review on Artificial Intelligence for Healthcare}

   \vspace{0.4cm}
   \large
   XAI


   \vspace{0.9cm}
     \textbf{Abstract}
\end{center}
%\lipsum[1]

Artificial Intelligence models are increasingly finding applications in the field of medicine.
Concerns have been raised about the explainability of the decisions that are made by these
AI models.In this article ,we have a systematic analysis of explainable artificial
intelligence(XAI),with a primary focus on models that are currently being used in the field of
health care.The literature search is conducted following the preferred reporting items for
systematic reviews and meta-analyses standards for relevant work published from first
January 2012 to second February 2022.The review analyses the prevailing trends in XAI and
lays out the major directions in which research is headed.We investigate the why,how and
when of the uses of these XAI models and their implications.We present a comprehensive
examination of XAI methodologies as well as an explanation of how a trustworthy AI can be
derived from describing AI models for health care fields.The discussion of the work will
contribute to the formalization of the XAI field.\\

XAI is emerging in various sectors and the literature has recognized numerous concerns and
problems.Security ,performance,vocabulary,evaluation of explanation and generalization of
XAI in the field of health care are the most significant problem recognized and discussed in
the literature.The paper has identified a number of obstacles associated with XAI in the field
of health care area including system evaluation,organizational,legal,socio-relational and
communication issues.Although a few studies in the literature have proposed some answers
for the stated problems,others have characterized it as a future gap that must be filled.In
health care,the organizational difficulty posed by XAI can be mitigated via AI deployment
management plans.\\

This paper contributes to the body of knowledge in identifying different techniques for XAI
in determining the issues and difficulties associated with XAI and finally in reviewing XAI in
health care.\\

\textbf{Keywords:}
XAI,AI,Security,health care


\end{document}
```

OUTPUT:

A Review on Artificial Intelligence for Healthcare

XAI

Abstract

Artificial Intelligence models are increasingly finding applications in the field of medicine. Concerns have been raised about the explainability of the decisions that are made by these AI models.In this article ,we have a systematic analysis of explainable artificial intelligence(XAI),with a primary focus on models that are currently being used in the field of health care.The literature search is conducted following the preferred reporting items for systematic reviews and meta-analyses standards for relevant work published from first January 2012 to second February 2022.The review analyses the prevailing trends in XAI and lays out the major directions in which research is headed.We investigate the why,how and when of the uses of these XAI models and their implications.We present a comprehensive examination of XAI methodologies as well as an explanation of how a trustworthy AI can be derived from describing AI models for health care fields.The discussion of the work will contribute to the formalization of the XAI field.

XAI is emerging in various sectors and the literature has recognized numerous concerns and problems.Security ,performance,vocabulary,evaluation of explanation and generalization of XAI in the field of health care are the most significant problem recognized and discussed in the literature.The paper has identified a number of obstacles associated with XAI in the field of health care area including system evaluation,organizational,legal,socio-relational and communication issues.Although a few studies in the literature have proposed some answers for the stated problems,others have characterized it as a future gap that must be filled.In health care,the organizational difficulty posed by XAI can be mitigated via AI deployment management plans.

This paper contributes to the body of knowledge in identifying different techniques for XAI in determining the issues and difficulties associated with XAI and finally in reviewing XAI in health care.

Keywords: XAI,AI,Security,health care

[Type here]

Experiment 3:

Develop a LaTeX script to create a simple title page of the VTU project Report [Use suitable Logos and text formatting]

```
\documentclass[12pt,a4paper]{article}
\usepackage{graphicx}
\usepackage{geometry}
\geometry{a4paper,total={180mm,250mm},left=20mm,top=30mm}
\thispagestyle{empty}
\begin{document}
\begin{center}
\large\textbf{VISVESVARAYA TECHNOLOGICAL UNIVERSITY}\\
Jnana Sangama,Belagavi-590018\\
\vspace{0.2in}
\includegraphics[scale=0.8]{vtu_logo.jpg}\\
\vspace{0.1in}
A PROJECT REPORT\\
ON\\
\textbf{"BIOMETRIC ATTENDANCE SYSTEM"}\\
\vspace{0.1in}
\textit{Submitted in partial fulfilment of the requirement for the award of the degree of}\\
\vspace{0.2in}
\textbf{BACHELOR OF ENGINEERING}\\
\textbf{IN}\\
\textbf{COMPUTER SCIENCE ENGINEERING}\\
\vspace{0.2in}
Submitted by
\begin{tabular}{p{12cm} p{13cm}}
\hspace{0.2cm} Anvitha S &  1TJ23CS001\\
\hspace{0.2cm} Akash R  & 1TJ23CS002\\
\hspace{0.2cm} Mansi K  & 1TJ23CS017\\
\hspace{0.2cm} Vanshika S  & 1TJ23CS025\\
\vspace{0.2in}
\end{tabular}
\textbf{Under the Guidance of }\\
Dr.S.Sajithra Varun\\
Associate Professor\\
Department of CSE\\
TJIT,Bangalore\\
\vspace{0.2in}
\includegraphics[scale=0.1]{logo.jpg}\\
DEPARTMENT OF CSE\\
\textbf{\large{T John Institute of Technology}}\\
Bangalore-560083\\
2024-2025\\

\end{center}
\end{document}
```

OUTPUT:

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama,Belagavi-590018

A PROJECT REPORT

ON

”BIOMETRIC ATTENDANCE SYSTEM”

Submitted in partial fulfilment of the requirement for the award of the degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE ENGINEERING

Submitted by

Anvitha S	1TJ23CS001
Akash R	1TJ23CS002
Mansi K	1TJ23CS017
Vanshika S	1TJ23CS025

Under the Guidance of

Dr.S.Sajithra Varun

Associate Professor

Department of CSE

TJIT,Bangalore

T.JOHN

INSTITUTE OF TECHNOLOGY

DEPARTMENT OF CSE

T John Institute of Technology

Bangalore-560083

2024-2025